

Effects of EPX on *A. caliginosa*

Protocol

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Objective : Construction of dose-response curves for growth and reproduction.

1 Generalities

1.1 Earthworms

Earthworms for these preliminary tests are *A. caliginosa* individuals from (or bred from individuals) fetch in a meadow in Versailles (Les Closeaux).

1.2 Design

Cosms are made with 200 g of Closeaux soil and earthworms are fed with 3 g/ind/14d.

Each modality is made with 5 replicates for growth and 5 replicates for reproduction.

The growth test lasts 28 days and the reproduction test lasts 28 days with a monitoring of cocoons of 28 days after the test.

For the growth test, one replicate correspond to 1 individuals per cosm. For the reproduction test, one replicate correspond to 2 individuals per cosm.

Table 1.1: Tested concentrations (mg/kg)

Tested concentrations (mg/kg)
0.0000
0.9283
2.0000
4.3089
9.2832
20.0000
43.0887
92.8318
200.0000

We need :

- 45 juveniles for the growth experiment,
- 90 adultes for the growth experiment.

1.3 Soil and horse dung needs

We need :

- 18 kg of Closeaux soil,
- 810 g of horse dung.

2 Protocol

2.1 Calculations

Table 2.1: Soil preparation for each EPX concentration

Concentration (mg/kg)	Q_{EPX} (mg)	Q_{tot} soil (g)	V_{eau} (CRE 70%)	$Q_{prim,dry}$ (g)	$Q_{prim,wet}$ (g)	Q_{nc} (g)	$V_{eau,nc}$ to 20% (mL)	V_{eau} 20 to 70% (mL)
0.000000	0.000000	2250	616.455	0.0000	0.000000	2250.000	176.1300	440.325
0.928318	2.088715	2250	616.455	6.9624	7.507399	2243.038	175.5850	440.325
2.000000	4.500000	2250	616.455	15.0000	16.174200	2235.000	174.9558	440.325
4.308869	9.694956	2250	616.455	32.3165	34.846258	2217.683	173.6003	440.325
9.283178	20.887150	2250	616.455	69.6238	75.073986	2180.376	170.6798	440.325
20.000000	45.000000	2250	616.455	150.0000	161.742000	2100.000	164.3880	440.325
43.088694	96.949561	2250	616.455	323.1652	348.462576	1926.835	150.8326	440.325
92.831777	208.871498	2250	616.455	696.2383	750.739861	1553.762	121.6285	440.325
200.000000	450.000000	2250	616.455	1500.0000	1617.420000	750.000	58.7100	440.325

2793.3062647 mL of primary solution n°1 are needed (837.9918794 mg of EPX).

2.2 D-1 - Preparation of chemical, soils and horse dung

Preparation of the primary soil :

- 870 mg of EPX are weighted in an inox recipient,
- 2900 g of soil are weighted in an another inox recipient,
- 227 mL of volvic is added to the soil and mixed,
- This soil preparation is gradually added and mixed to the inox recipient containing EPX.

Preparation of soils : (water used = Volvic)

- For the controls :
 - 2250 g of soil is weighted in an inox recipient,
 - 616.5 mL of water are added (do not mix),

- The recipient is covered with aluminum.
- For each concentration
 - The corresponding Q_{nc} g of dry soil is weighted in an big inox recipient,
 - The corresponding $V_{eau,nc}$ to 20% mL of water is added,
 - The soil is mixed.
 - The corresponding $Q_{prim,wet}$ is weighted,
 - The soil is mixed for at least 10 min,
 - 1213.065 g of soil are weighted in two smaller inox recipient,
 - Half of V_{eau} 20 to 70% is added in each recipient (do not mix).

Preparation of horse dung :

- 450 g of dry horse dung are prepared with 1590 mL of water,
- 45×13.6 g of this preparation is weighted in small boxes for the growth cosms,
- 45×27.2 g of this preparation is weighted in small boxes for the reproduction cosms.

2.3 D0 - Preparation of cosms and earthworms introduction

For each concentration :

- Recipient 1 and recipient 2 are mixed individually and then together for at least 10 min,
- This preparation is used to prepare 10 cosms (254.8 g),
- The rest of this preparation is put into sample pots that are kept in a thermoregulated chamber at -20°C ,
- In 5 cosm (growth), 13.6 g of horse dung is added and mixed,
- In 5 cosm (reproduction), 27.2 g of horse dung is added and mixed.

Earthworms introduction :

- For growth, earthworms are weighted individually and put in their respective cosms,
- For reproduction, earthworms are weighted in groups of 2 and put in their respective cosms,
- Cosms are weighted and their weight is written on their labels to be able to follow the soil humidity,
- Cosms are kept in a thermoregulated enclosure at 18°C .

2.4 Maintenance

On D7, D14 and D21, soil humidity is checked.

On D14, earthworms are fed and weighted the same way as D0. Cosms are weighted again.

2.5 D28 - End of exposition

On D28, earthworms are weighted and put on wet filter paper in Petri dishes. Soil samples are taken for each concentration in 125 mL recipients.

2.6 D31 - Freeze of individuals and cocoons recovery

Earthworms are weighted and put in test tubes in a -80°C refrigerator.

2.7 D31-56 - Cocoons

Each day of the working week, cocoons eclosions are checked.